# HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

## PROFESSIONAL EXPERIENCE — 3.5 YEARS

**Software Engineer — HCL Technologies, Chennai, India** Sep 2021 – Jan 2025

- Deployed and configured **FabGuard monitoring systems** for Etch and CVD tool groups, ensuring seamless integration with factory operations.
- Provided on-site technical support for **FabGuard software and hardware**, diagnosing and resolving issues to minimize downtime for semiconductor manufacturing processes.
- Performed PC installation, configuration, and cabling for new deployments, ensuring reliable data acquisition infrastructure.
- Developed **data acquisition parameters** and supported integration projects using industrial protocols such as **OPC-UA** and **SECS/GEM**, enabling robust system connectivity.
- Tested and validated **software updates** for FabGuard systems, troubleshooting connectivity issues and data discrepancies to maintain system integrity.
- Contributed to process improvement initiatives by analyzing FabGuard data and identifying opportunities for optimization in fault detection and classification.
- Documented site-specific configurations and conducted knowledge transfer sessions with customer teams on **FabGuard procedures** and best practices.

## SKILLS

**Monitoring Systems** – FabGuard, Data Acquisition, Fault Detection and Classification (FDC)

**Industrial Protocols** – OPC-UA, SECS/GEM, TCP/IP

**Operating Systems** – Windows (Installation, Configuration, Troubleshooting)

**Software Development Lifecycle** – Testing, Validation, Troubleshooting, Software Updates

**Customer Interaction** – On-site Support, Technical Training, Feedback Provision, Site Coordination

**Manufacturing Processes** – Semiconductor Manufacturing, HVAC, Automotive Manufacturing, Thin-film Coatings, Vacuum Processes

## PROJECTS

### FabGuard System Deployment and Configuration

- Engineered the deployment and configuration of the **FabGuard monitoring system** across multiple semiconductor fabrication tools, including Etch and CVD.
- Managed on-site installation and troubleshooting of **FabGuard hardware and software**, resolving complex technical issues to ensure operational readiness.
- Provided comprehensive customer training on system operation and data analysis, empowering teams to optimize production quality.

### Industrial Protocol Integration for Data Acquisition

- Integrated **FabGuard systems** with existing manufacturing equipment using industrial protocols such as **OPC-UA** and **SECS/GEM**.
- Developed and validated data acquisition parameters, ensuring accurate and reliable data flow for real-time monitoring and analysis.
- Troubleshot and resolved data discrepancies and connectivity issues, ensuring the integrity of the acquired manufacturing data.

## EDUCATION

**M.Sc. Data Analytics — National College of Ireland, Dublin** Feb 2026

**B.E. Computer Science — SNS College of Technology, Coimbatore, India** 2021

## CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)